

- Q.22 Fall arrestor
 Q.23 Improving safety and productivity through work place design
 Q.24 Role of preventive maintenance in accident prevention.
 Q.25 Safe use of various types of hand tools used for metal cutting
 Q.26 State the safety features associated with design, construction and use of scaffolds of different types.
 Q.27 Competent Person as per factories Act, 1948.
 Q.28 Safe use and maintenance of wire rope slings.
 Q.29 Safety majors while working on roofs.
 Q.30 Safety features associated with stairways.
 Q.31 Safe storage of materials.
 Q.32 Potential tripping and slipping hazards.
 Q.33 Principles of machine guarding, including the type machine guards.
 Q.34 Design a permit system for Height Work
 Q.35 Maintenance and repair of guards

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
 Q.36 What do you mean by HAVS? Write in detail the causes for developing HAVS and Suggest remedial measures to prevent the same.
 Q.37 Write down short note
 i) Team lifting and carrying non sparking tools
 ii) Safe use of accessories for manual handling
 Q.38 Discuss about selection, inspection, maintenance, repairs and safe use of Portable Power tools.

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1st Year / Industrial Safety Subject:- Safety Engineering - I

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 What is the primary purpose of machine guarding?
 a) Enhancing the appearance of the machine
 b) Preventing unauthorized access to the machine
 c) Protecting workers from hazardous machine parts
 d) Making the machine more efficient
 Q.2 What is Lockout / Tagout (LOTO) used for?
 a) Identifying the owner of the machine
 b) Preventing the machine from being used by unauthorized personnel
 c) Ensuring proper machine maintenance and repair
 d) Keeping machines running smoothly
 Q.3 What is covered under material handling?
 a) Lifting b) Loading
 c) Packing / unpacking d) Storage
 e) All of the above
 Q.4 Load Marking are posted on
 a) All sides of the crane and on the hoist block
 b) Workshop notice board
 c) Main gate of workshop
 d) Maintenance Register
 Q.5 What should you never do when carrying a load in forklifts?

- a) Adjust forks so that the load is 6 to 10 inches above the ground
 - b) Drive the forklift slowly and with clear vision of the pathway
 - c) Keep forks tilted back
 - d) Carry some light load on the overhead guard
- Q.6 What are the three Fall Prevention Systems for work at height?
- a) Full body harness, anchorage and connectors
 - b) Top rail, mid rail, and toe-board
 - c) Safety net, harness and lifeline
 - d) None of the above
- Q.7 Before installing a new abrasive wheel on a portable hand-held grinder, you are required to:
- a) inspect wheel for cracks and damage
 - b) check RPM rating to ensure it is right for the grinder
 - c) Perform a "ring test" to determine if the wheel is cracked
 - d) all of the above
- Q.8 Which is not a part of preventive maintenance?
- a) Routine maintenance
 - b) Periodic maintenance
 - c) Overhaul
 - d) None of above
- Q.9 When working with power tools outdoors or in a damp environment, where should the cord be plugged in to?
- a) A non-waterproof extension cord
 - b) A GFCI (Ground Fault Circuit Interrupter)
 - c) An outlet without a grounding plug
 - d) All of the above
- Q.10 Section 25 of the Factories Act, 1948 contains provisions on _____

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- a) Employment of young person's on dangerous machines
- b) Striking gears and devices for cutting off power
- c) Self-acting machines
- d) Work on a near machinery in motion

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 The beam which is fixed at one end and free at other end is called _____
- Q.12 Frequency of testing and certifying chain slings by competent person as per Factories Act 1948, is _____ in a year.
- Q.13 What is the full form of EOT pertaining to crane?
- Q.14 Formula of Factor of Safety.
- Q.15 Vibration and radiation come under which type of hazard.
- Q.16 Full form of PASS
- Q.17 In many cases, wire rope is subjected to impact loads which send axial and torsional responses up and down the rope. If the loads are severe enough, the outer strands can separate from the core in a permanent manner and thus render the rope useless. Such a phenomenon is generally known as _____
- Q.18 While working near flammable substances, one should avoid iron or steel hand tools. (True/False)
- Q.19 As per OSHA, The safe exposure limits for noise levels for 08 hours of working/day is _____
- Q.20 An abrasive wheel is checked for cracks or defects with a _____

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Basic requirements of guards

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